

Lecture 1 Academic Style

♣ Academic Style Practice

Rewrite the sentences so they are more academic.

1. Another thing to think about is the chance of crime getting worse.
2. Regrettably these days lots of people don't have jobs.
3. Sometime soon they will find a vaccine for malaria.
4. A few years ago the price of property in Japan went down a lot.

♣ Technical Writing

Compare and contrast the language style of passage 5 and passage 6.

5. *What is GPS?*

GPS stands for the Global Positioning System. It refers to a system of satellites and receivers that allow people and devices to pinpoint their precise location on the earth. The heart of the system relies on 24 satellites that orbit the earth twice per day. Devices that are equipped with GPS equipment receive transmissions from at least a few of the satellites and are able to discern very precise positioning data.

(Quoted from: What is GPS? [Online]. Available: <http://www.wisegeek.com/what-is-gps.htm>)

6. *GPS Overview*

GPS consists, nominally, a constellation of 24 operational satellites. This constellation, which was completed in July 1993, was known as the initial operational capability (IOC). The official IOC announcement, however, was made on December 8, 1993. To ensure continuous worldwide coverage, GPS satellites are arranged so that four satellites are placed in each of six orbital planes (Figure 1.1). With this constellation geometry, four to ten GPS satellites will be visible anywhere in the world, if an elevation mask of 10 degrees is considered. Only four satellites are needed to provide the positioning, or location, information.

(Quoted from: A. El-Rabbany, Introduction to GPS: the Global Positioning System)

7. *Exoskeletons*

Mastery of the outstanding skills characterizing experts accompanies extensive practicing. However, practice does not always guarantee improvement. For trained individuals such as athletes and musicians, learning often plateaus after extensive training (1-4), known as the "ceiling effect". One bottleneck to overcome it is having no prior physical experience with the skill to be learned (5-6). Here, we challenge this issue by exposing expert pianists to fast and complex finger movements that cannot be performed voluntarily, using a hand exoskeleton robot that can move individual fingers quickly and independently. Although the skill of moving the fingers quickly plateaued through weeks of piano practice, passive exposure to otherwise impossible complex finger movement generate by the exoskeleton robot at a speed faster than the pianists fastest one enabled them to play faster. Neither a training for fast but simple finger movements nor one

for slow but complex movements with the exoskeleton enhanced the overtrained motor skill. The exoskeleton training with one hand also improved the motor skill of the untrained contralateral hand, demonstrating the intermanual transfer effect.

(Quoted from: Furuya et al., *Surmounting the ceiling effect of motor expertise by novel sensory experience with a hand exoskeleton*. *Science Robotics*, eadn3802, 2025.)

➤ Exercise

Deforestation

In the last decade, there has been a dramatic increase in deforestation, particularly 1. well-known in the Amazon rainforest. Inpe (2020) reports that between 2019 and 2020 it rose by almost 10%, making the deforestation rate of the Legal Amazon region in Brazil reached the highest level in 15 years. According to AAF (2020), one of the 2. biggest issues with deforestation is the impact on ground surface for more sunlight to penetrate the forest floor, causing it to dry out and to heat up. Consequently, overall temperatures rise, which 3. adds to the possibility of drought. Furthermore, once the area of vegetation 4. gets smaller, the soil erodes and decays, leaving many organisms 5. without nutrients. As a result, animals need to search in the more dangerous 6. openness to survive 7. less and less of space and food. 8. Getting rid of the trees also plays a crucial role in determining the climate. As the trees are burned, the greenhouse gas is 9. let out, which 10. then contributes to regional and global warming.

Please exchange the informal words underlined in the text with academic expressions below in proper forms.

derive	cause	wilderness	notable	shortage
exposure	significant	release	approach	remove
diminish	factor	devoid of	further	precise

1. _____ 2. _____ 3. _____ 4. _____ 5. _____

6. _____ 7. _____ 8. _____ 9. _____ 10. _____